week 8

Find the limit of $f(x)$ as x approaches 2, where f is defined as

$$f(x) = \begin{cases} 1, & x \neq 2 \\ 0, & x = 2 \end{cases}.$$

EXAMPLE 1 Evaluating Basic Limits

a. $\lim_{x \to 2} 3 =$ **b.** $\lim_{x \to -4} x =$ **c.** $\lim_{x \to 2} x^2 = 2^2 =$

$$\lim_{x \to 2} (4x^2 + 3)$$

$$\lim_{x \to 1} \frac{x^2 + x + 2}{x + 1}.$$

$$\lim_{x \to 0} \sqrt{x^2 + 4}$$

$$\lim_{x \to 3} \sqrt[3]{2x^2 - 10}$$

a. $\lim_{x \to 0} \tan x =$

b. $\lim_{x \to \pi} (x \cos x) =$

c. $\lim_{x \to 0} \sin^2 x =$

$$\lim_{x \to 1} \frac{x^3 - 1}{x - 1}.$$

$$\lim_{x \to -3} \frac{x^2 + x - 6}{x + 3}.$$

$$\lim_{x \to 0} \frac{\sqrt{x + 1} - 1}{x}.$$

Determine if the following function is continuous at *x*=2 .

$$f(x)=\begin{cases} x^2+2x, & if \quad x\le 2 \\ x^3-6x, & if \quad x>2 \end{cases}$$

Find the limit of $f(x) = \sqrt{4 - x^2}$ as x approaches -2 from the right.

Discuss the continuity of $f(x) = \sqrt{1 - x^2}$.

$\geqslant$ $\quad$ $y \geqslant 1$

1

11. $\lim_{x \to 6} (x - 2)^2$

12. $\lim_{x \to 7} (10 - x)^4$

13. $\lim_{t \to 4} \sqrt{t + 2}$

14. $\lim_{y \to 4} 3|y - 1|$

15. $\lim_{t \to -2} \dfrac{t + 2}{t^2 - 4}$

16. $\lim_{t \to 3} \dfrac{t^2 - 9}{t - 3}$

17. $\lim_{x \to 4} \dfrac{\sqrt{x - 3} - 1}{x - 4}$

18. $\lim_{x \to 0} \dfrac{\sqrt{4 + x} - 2}{x}$

week 9

Find the slope of the graph of

$$f(x) = 2x - 3$$

at the point (2, 1).

2

Find the slopes of the tangent lines to the graph of

$$f(x) = x^2 + 1$$

2x

at the points (0, 1) and (−1, 2), as shown in Figure

Find the derivative of $f(x) = x^3 + 2x$.

Find $f'(x)$ for $f(x) = \sqrt{x}$. Then find the slopes of the graph of f at the points $(1, 1)$ and $(4, 2)$. Discuss the behavior of f at $(0, 0)$.

ถ้า function ไม่ต่อเนื่องที่จุดนั้น = ไม่สามารถหา diff ณ จุดนั้นได้

แต่ถ้าบอกว่า function ต่อเนื่องที่จุดนั้น ไม่ได้หมาย ค.ว่า สามารถหา diff ณ จุดนั้นได้

The function

$$f(x) = |x - 2|$$

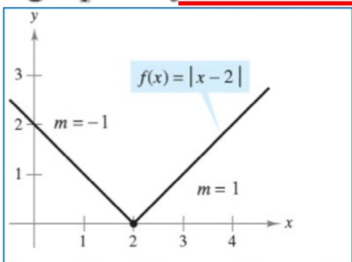
is continuous at $x = 2$. However, the one-sided limits

$$\lim_{x \to 2^-} \frac{f(x) - f(2)}{x - 2} = \lim_{x \to 2^-} \frac{|x - 2| - 0}{x - 2} = -1 \qquad \text{Derivative from the left}$$

and

$$\lim_{x \to 2^+} \frac{f(x) - f(2)}{x - 2} = \lim_{x \to 2^+} \frac{|x - 2| - 0}{x - 2} = 1 \qquad \text{Derivative from the right}$$

are not equal. So, f is not differentiable at $x = 2$ and the graph of f does not have a tangent line at the point $(2, 0)$.



f is not differentiable at $x = 2$, because the derivatives from the left and from the right are not equal.

Function
a. $y = 7$
b. $f(x) = 0$
c. $s(t) = -3$
d. $y = k\pi^2$, k is constant

Function
a. $f(x) = x^3$
b. $g(x) = \sqrt[3]{x}$
c. $y = \dfrac{1}{x^2}$

Find the slope of the graph of $f(x) = x^4$ when $4x^3$

a. $x = -1$ **b.** $x = 0$ **c.** $x = 1$.

Find an equation of the tangent line to the graph of $f(x) = x^2$ when $x = -2$.

Function

a. $y = \dfrac{2}{x}$

b. $f(t) = \dfrac{4t^2}{5}$

c. $y = 2\sqrt{x}$

d. $y = \dfrac{1}{2\sqrt[3]{x^2}}$

e. $y = -\dfrac{3x}{2}$

Original Function

a. $y = \dfrac{5}{2x^3}$

b. $y = \dfrac{5}{(2x)^3}$

c. $y = \dfrac{7}{3x^{-2}}$

d. $y = \dfrac{7}{(3x)^{-2}}$

Function
a. $f(x) = x^3 - 4x + 5$
b. $g(x) = -\dfrac{x^4}{2} + 3x^3 - 2x$

Function
a. $y = 2\sin x$
b. $y = \dfrac{\sin x}{2} = \dfrac{1}{2}\sin x$
c. $y = x + \cos x$

Using the Product Rule

Find the derivative of $h(x) = (3x - 2x^2)(5 + 4x)$.

Find the derivative of $y = 3x^2 \sin x$.

Find the derivative of $y = 2x \cos x - 2 \sin x$.

Find the derivative of $y = \dfrac{5x - 2}{x^2 + 1}$.

Original Function

a. $y = \dfrac{x^2 + 3x}{6}$

b. $y = \dfrac{5x^4}{8}$

c. $y = \dfrac{-3(3x - 2x^2)}{7x}$

d. $y = \dfrac{9}{5x^2}$

Differentiate both forms of $y = \dfrac{1 - \cos x}{\sin x} = \csc x - \cot x.$

Solution

First form: $y = \dfrac{1 - \cos x}{\sin x}$

$$y' = \frac{(\sin x)(\sin x) - (1 - \cos x)(\cos x)}{\sin^2 x}$$

$$= \frac{\sin^2 x + \cos^2 x - \cos x}{\sin^2 x}$$

$$= \frac{1 - \cos x}{\sin^2 x}$$

Second form: $y = \csc x - \cot x$

$$y' = -\csc x \cot x + \csc^2 x$$

To show that the two derivatives are equal, you can write

$$\frac{1 - \cos x}{\sin^2 x} = \frac{1}{\sin^2 x} - \left(\frac{1}{\sin x}\right)\left(\frac{\cos x}{\sin x}\right)$$

$$= \csc^2 x - \csc x \cot x.$$

$$g(x) = (x^3 + 7x)(x + 3)$$

$$\mathrm{f}(x) = \frac{6x-5}{x^2+1}$$

$$\mathrm{y} = \frac{\sin x}{x^4}$$

week 10

$$y = \sin x \qquad\qquad y = \sin 6x$$

$$y = 3x + 2 \qquad\qquad y = (3x + 2)^5$$

$$y = x + \tan x \qquad\qquad y = x + \tan x^2$$

Without the Chain Rule	*With the Chain Rule*
$y = x^2 + 1$	$y = \sqrt{x^2 + 1}$
$y = \sin x$	$y = \sin 6x$
$y = 3x + 2$	$y = (3x + 2)^5$
$y = x + \tan x$	$y = x + \tan x^2$

$y = f(g(x))$

a. $y = \dfrac{1}{x + 1}$

b. $y = \sin 2x$

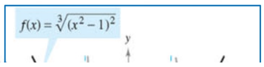
c. $y = \sqrt{3x^2 - x + 1}$

d. $y = \tan^2 x$

Find the derivative of $f(x) = (3x - 2x^2)^3$.

$$y = (x^2 + 1)^3.$$

Find all points on the graph of $f(x) = \sqrt[3]{(x^2 - 1)^2}$ for which $f'(x) = 0$ and those for which $f'(x)$ does not exist.



Differentiate $g(t) = \dfrac{-7}{(2t - 3)^2}$.

$$y = \left(\frac{3x - 1}{x^2 + 3}\right)^2$$

$$y = \left(\frac{3x - 1}{x^2 + 3}\right)^2$$

shows.

$$\frac{d}{dx}[\sin u] = (\cos u)\, u' \qquad \frac{d}{dx}[\cos u] = -(\sin u)\, u'$$

$$\frac{d}{dx}[\tan u] = (\sec^2 u)\, u' \qquad \frac{d}{dx}[\cot u] = -(\csc^2 u)\, u'$$

$$\frac{d}{dx}[\sec u] = (\sec u \tan u)\, u' \qquad \frac{d}{dx}[\csc u] = -(\csc u \cot u)\, u'$$

a. $y = \sin 2x$

b. $y = \cos(x - 1)$

c. $y = \tan 3x$

a. $y = \cos 3x^2 = \cos(3x^2)$

b. $y = (\cos 3)x^2$

c. $y = \cos(3x)^2 = \cos(9x^2)$

d. $y = \cos^2 x = (\cos x)^2$

e. $y = \sqrt{\cos x} = (\cos x)^{1/2}$

a. $y = \cos 3x^2 = \cos(3x^2)$

b. $y = (\cos 3)x^2$

c. $y = \cos(3x)^2 = \cos(9x^2)$

d. $y = \cos^2 x = (\cos x)^2$

e. $y = \sqrt{\cos x} = (\cos x)^{1/2}$

$$f(t) = \sin^3 4t$$

a. $\dfrac{d}{dx}[\ln(2x)] = \dfrac{u'}{u} =$

b. $\dfrac{d}{dx}[\ln(x^2 + 1)] =$

c. $\dfrac{d}{dx}[x \ln x] =$

$=$

d. $\dfrac{d}{dx}[(\ln x)^3] =$

$=$

$$f(x) = \ln|\cos x|.$$

a. $\dfrac{d}{dx}\left[e^{2x-1}\right] =$

b. $\dfrac{d}{dx}\left[e^{-3/x}\right] =$

Math f

Find dy/dx given that $\quad x^2 + 4y^2 = 4$

Find dy/dx given that $y^3 + y^2 - 5y - x^2 = -4$.

Find dy/dx given that $x^4 + x^2y^2 - y^2 = 0$

Given $x^2 + y^2 = 25$, find $\dfrac{d^2y}{dx^2}$. Evaluate the first and second derivatives at the point $(-3, 4)$.

$$y^2 = (x - y)(x^2 + y)$$

$$y\sqrt{x} - x\sqrt{y} = 25$$

$$\cos(x + y) = x$$

Week 11

EXAMPLE 3 Finding Extrema on a Closed Interval

Find the extrema of $f(x) = 2x - 3x^{2/3}$ on the interval $[-1, 3]$.

Solution Begin by differentiating the function.

EXAMPLE 4 Applying the First Derivative Test

Find the relative extrema of $f(x) = \dfrac{x^4 + 1}{x^2}$.

EXAMPLE Evaluate $\lim_{x \to -\infty} \frac{x^2}{e^{-x}}$

Week 12

EXAMPLE Evaluate $\lim_{x \to 0} \frac{e^{2x} - 1}{x}$.

Basic Integration Rules

EXAMPLE

$$\boxed{\int k\,dx = kx + C}$$

$$\boxed{\int kf(x)\,dx = k\int f(x)\,dx}$$

$$\boxed{\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,}$$

$\int 5dx$

$\int -8dx$

$\int \beta dx$

$\int -2xdx$

$\int \pi x dx$

Basic Integration Rules

EXAMPLE

$\int x^2 dx$

$\int x^5 dx$

$\int x^9 dx$

$\int 3x^2 dx$

$\int 10x^4 dx$

$\int 8x^7 dx$

Basic Integration Rules

$$\int \sqrt[4]{x}dx$$

$$\int \frac{1}{x} dx$$

$$\int \frac{2}{x} dx$$

$$\int \frac{5}{x} dx$$

Basic Integration Rules

$$\int \frac{1}{x^5}\,dx$$

$$\int \frac{1}{\sqrt{x}}\,dx$$

$$\int \frac{1}{\sqrt[3]{x}}\,dx$$

EXAMPLE 6 Rewriting Before Integrating

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \frac{\sin x}{\cos^2 x} \, dx$$

$$\int 3 \cos x \, dx$$

$$\int \frac{1}{\sin^2 x} \, dx$$

$2x^4 + 3x^{\frac{5}{3}} + 2x^{-5} + 7x +$

1) $f'(x) = 8x^3 + 5\sqrt[3]{x^2} - \frac{10}{x^6} + 7$

2) $g'(x) = \frac{3\cos x}{\sin^2 x}$

3) $y' = \frac{x^2+3\sqrt{x}}{x^3}$

Week 13

EXAMPLE 2 **A Definite I**

$\gtrless 0$

Evaluate $\displaystyle\int_0^2 |2x - 1|\, dx.$

EXAMPLE 3 Using the Fundamental Theorem to Find Area

Find the area of the region bounded by the graph of $y = 2x^2 - 3x + 2$, the x-axis, and the vertical lines $x = 0$ and $x = 2$, as shown in Figure

EXAMPLE 4 Finding the Average Value of a Function

Find the average value of $f(x) = 3x^2 - 2x$ on the interval $[1, 4]$.

EXAMPLE 4 Change of Variables

Find $\int \sqrt{2x - 1}\, dx$.

EXAMPLE 5 Change of Variables

Find $\int x\sqrt{2x-1}\,dx$. $\frac{du}{dx} = 2 \rightarrow$

a. $\int 3(3x - 1)^4 \, dx = \int (3$

b. $\int (2x + 1)(x^2 + x) \, dx$

c. $\int 3x^2 \sqrt{x^3 - 2} \, dx = \int ($

d. $\int \frac{-4x}{(1 - 2x^2)^2} \, dx = \int$

e. $\int \cos^2 x \sin x \, dx = -\int$

EXAMPLE

$$\int (2x + 3)^7 \, dx$$

EXAMPLE

$$\int x(x^2 + 1)^4 dx$$

EXAMPLE

$$\int 3x^2\sqrt{x^3+5}\,dx$$

EXAMPLE

$$\int \frac{x}{3x^2+4}\,dx$$

EXAMPLE

$$\int \sin(3x + 2)\, dx$$

Integration by

EXAMPLE $\int \sin x \cos x \, dx$

EXAMPLE 9 Change of Variables

Evaluate $A = \int_1^5 \frac{x}{\sqrt{2x - 1}}\, dx$.

$$\int_{-1}^{1}\left(6x^5-\frac{2}{\sqrt[5]{x^3}}+3\right)dx$$

$$\int_{1}^{4}|x-3|\,dx$$

$$\int\frac{(2x^2+1)}{(2x^3+3x)^2}\,dx$$

$$\int_{-1}^{1}\left(6x^5 - \frac{2}{\sqrt[5]{x^3}} + 3\right)dx$$

$$\int_{1}^{4}|x-3|\,dx$$

$$\int \frac{(2x^2+1)}{(2x^3+3x)^2}\,dx$$